

BHANNAK ABHILASH

+91 8309601506 | bhannakabhilash@gmail.com | [linkedin.com/in/bhannakabhilash/](https://www.linkedin.com/in/bhannakabhilash/)

Education

B V Raju Institute of Technology

B.Tech in Electronics and Communication Engineering — CGPA: 8.82

Oct 2022 – May 2026

Narsapur, Medak

SV Junior College

Intermediate in Mathematics, Physics & Chemistry — Marks: 974

Aug 2020 – May 2022

Sangareddy

Sai Krishnaveni High School

Secondary School Certificate — CGPA: 10

April 2020

Jogipet

Technical Skills

Programming Languages: Python, C, SQL, Verilog, and System Verilog

Frameworks & Technologies: FastAPI, LangChain, Generative AI, Retrieval-Augmented Generation (RAG), Prompt Engineering

Python Libraries: NumPy, Pandas

Developer Tools: Git, VS Code, Vivado, MATLAB

Hardware: FPGA (PYNQ Board)

Experience

GenAI & Python Developer Trainee

Trysol Global Services Pvt. Ltd.

Project: Student Assistant Chatbot

Technologies: Python, FastAPI, RAG, LLMs, LangChain, Git, Prompt Engineering

- Built a **Student Assistant Chatbot** using Retrieval-Augmented Generation (RAG) to answer student queries intelligently.
- Loaded and processed PDF documents containing student details and academic resources as the knowledge base for the chatbot.
- Developed and exposed chatbot functionality as REST API endpoints using **FastAPI** for seamless integration.
- Implemented document parsing and vector-based retrieval pipeline to enable accurate, context-aware responses.
- Managed version control and pushed project codebase to remote repository using **Git**.
- Applied prompt engineering techniques to improve LLM response quality and relevance.

Competitions & Hackathons

Digital Twin for Battery Management System — L&T Techgium 2025 (Runner Up) Feb – Apr 2025

Technologies: Python, LSTM, Machine Learning, BMS, Data Analysis

- Built a Digital Twin to eliminate sensor inefficiencies and improve battery health monitoring.
- Achieved 95% accuracy between Digital Twin and hardware data using key battery parameters.
- Trained an LSTM model on a custom dataset for accurate State of Charge (SOC) prediction.
- Created an Enhanced Mode that fuses real and simulated data for more reliable BMS outputs.

Smart India Hackathon (SIH) — Wearable Safety Device for Women

Aug 2024

- Discreet panic button triggering GPS, audio, and video capture for emergencies.
- Automatic alert system designed to significantly reduce emergency response time.
- Low-power mode ensures battery efficiency until device activation.

Academic Projects

Augmentative and Alternative Communication (AAC) System

Jun – Aug 2024

Technologies: Python, FPGA

- Developed an AAC system to assist users with speech or motor impairments in daily communication.
- Ensured multi-input accessibility (keyboard, touch and mouse) with real-time response optimization.

Multi-Metric Analysis of Power Efficient Digital Circuits

Apr 2025 – Mar 2026

Technologies: Verilog, CMOS Logic, Adiabatic Logic, Python, Data Analysis

- Designed adiabatic logic-based digital circuits in Verilog to enhance power efficiency over conventional CMOS.
- Simulated circuits in Vivado and validated significant power savings through energy recovery.
- Built a Python dashboard integrating CSV exports from Vivado for streamlined data analysis.

Certifications

- Smart Interviews: Smart Coder
- LetsUpgrade: Python Bootcamp
- NPTEL: System Design Through Verilog
- Udemy: SystemVerilog (SV) Language